



Customer Segmentation with Clustering

Mini Project 6.3 - Report on Experiments & Results

By Mahmood Ahmad



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1. Problem Statement

Apply clustering techniques of data science, on a large e-commerce dataset from a real-world organisation, to perform customer segmentation that can assist improve marketing efforts.

2. Description of Provided Data set

The original dataset comprises of 20 variables with 951, 668 rows of customer orders from 47 different countries of the world.

Some null values were found in three variables, namely: 'City', 'Postal_Code' and 'State_Province'. However, these variables will not be included in the clustering approach and would later be dropped from our analysis.

3. Overview of the Approach

3.1 Insights from Data

Majority of features were not being in numeric format, dates of births for customers needed to be changed in a format which we could use in Python for our analysis. Some features like **City**, **Postal_code**, and **State_Province** had null values. However, as we were guided to use some other features for our analysis, we ignored these facts for these features. The currency-based features, namely **Total Revenue** and **Unit Cost** needed removing the \$ sign and the brackets around numbers and thus conversion from string to float, which we carried out successfully.

3.2 Feature Engineering

The feature engineering included adding five features, which were later be useful for our later analysis and customer segmentation: these were **Frequency** (derived from number of orders per customer), **Recency**, **CLV** (customers' lifetime value), **Avg_Unit_Cost** and **Age** (derived from customer's date of birth).

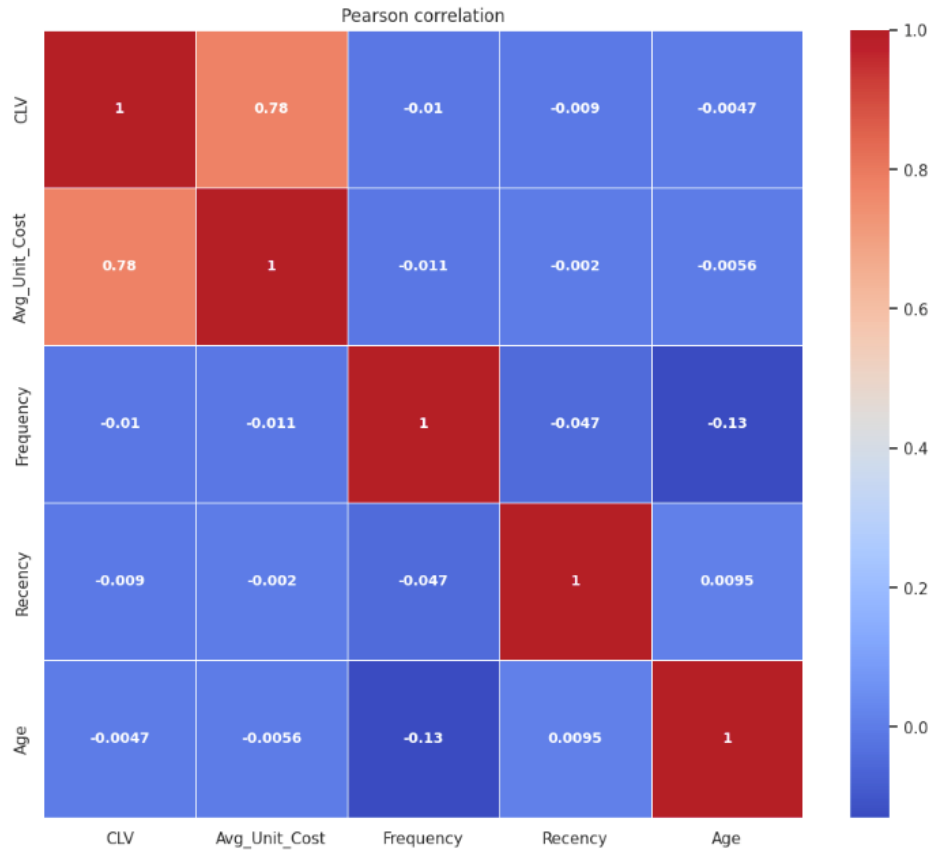
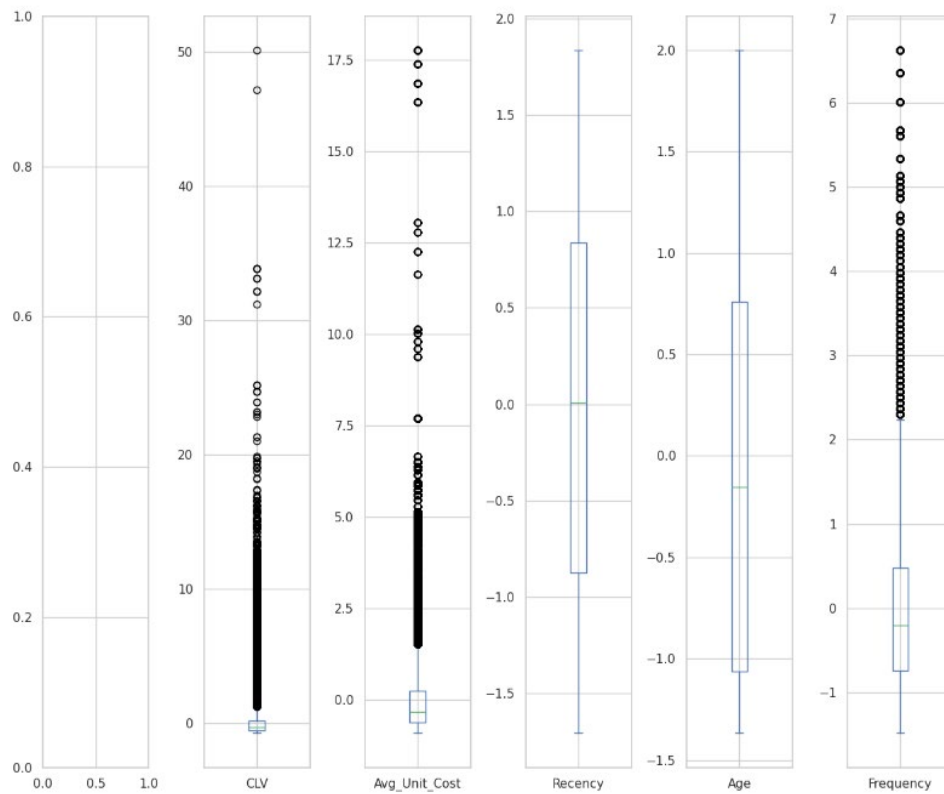


Figure 1: Seaborn's correlation heatmap of the five features selected for Clustering.



Identifying the duplicates and obtaining one row per customer in data decreased the data size from 951668 rows to only 68299 rows with five columns containing the features that

were considered significant for analysis.

4. Customer Segmentation

4.1 Selecting the optimal number k of Clusters - The Elbow Method

As the Elbow method identifies the point on the plot where the rate of decrease in within-cluster variation (wcss) starts to diminish, forming a distinct bend or elbow (Cardona, 2024), we found a low rate of decrease for WCSS at $n = 4$, and this seemed to be the optimum number of clusters for data of this size.

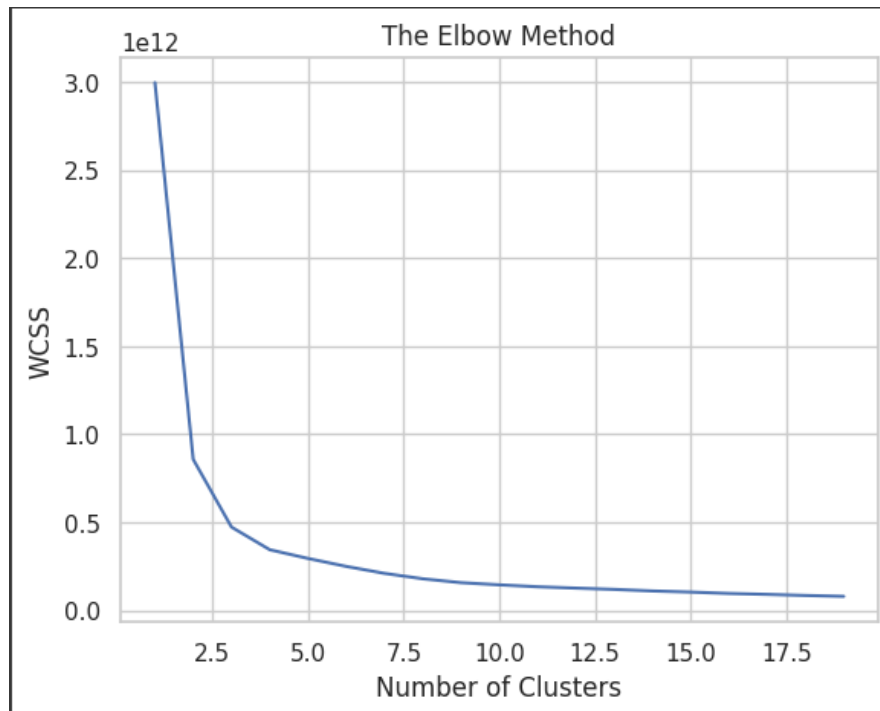


Figure 2: Result of applying the Elbow method to find optimum number of clusters, data size = 30,000.

4.2 The Silhouette Method

We applied the Silhouette method for varying number of clusters ($n_clusters$) from 2 to 8. Due to memory constraints. We computed Silhouette score for a sample of about 30000 observations (rows) of data, which gave the best Silhouette score for $n_clusters=2$ (Figure 3), with gradually decreasing to a smaller number for larger $n_cluster$ value. We decided to take $n_cluster=4$ as the optimal number of clusters for this data (Figure 2). This is because the data size is too large, and the business context suggests that we should follow the Elbow method.

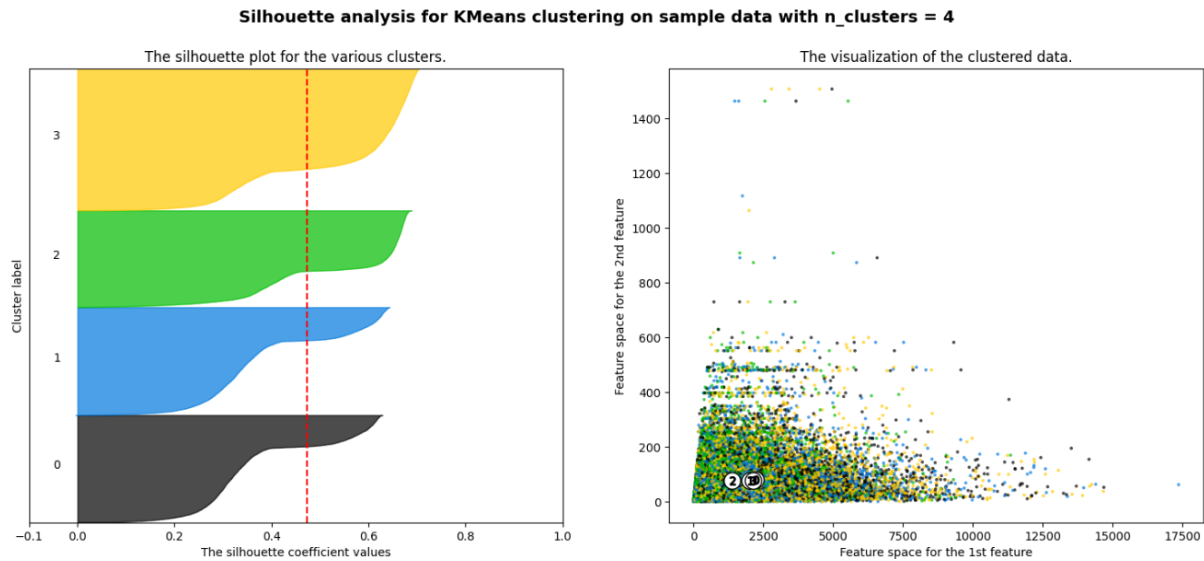


Figure 3: The Silhouette Plot for $n_clusters = 4$. Sample data size = 30000

4.3 The Hierarchical Clustering (Dendrogram) Method

Figure 4 is a 2D visualisation of hierarchical clustering, a considerable simplification from 5D data.

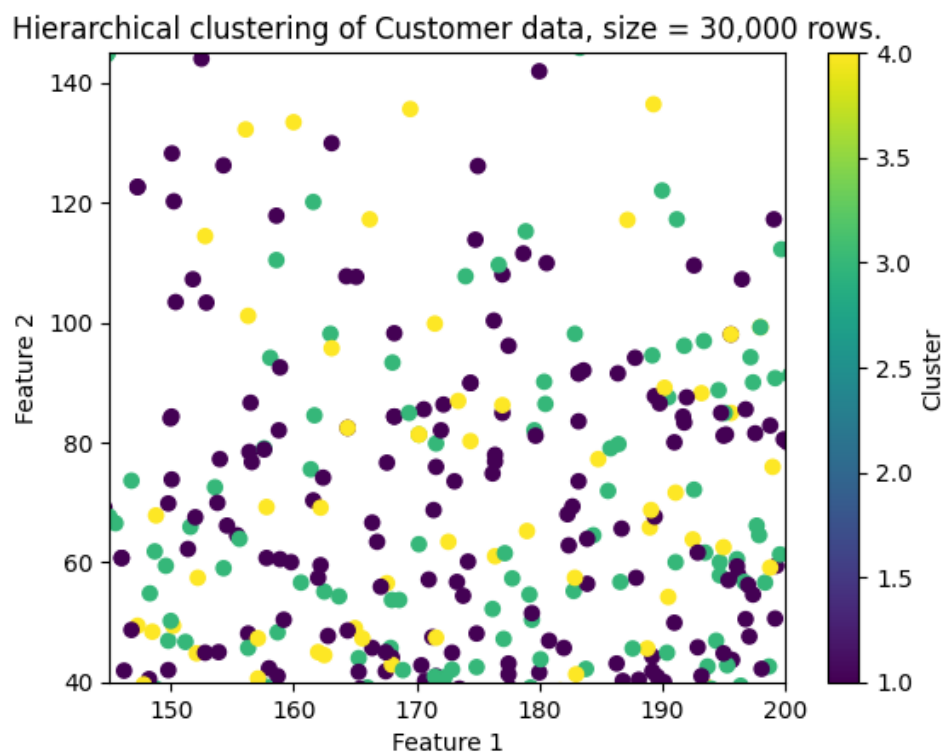


Figure 4: Hierarchical Clustering of Customer data of size = 30,000 rows.

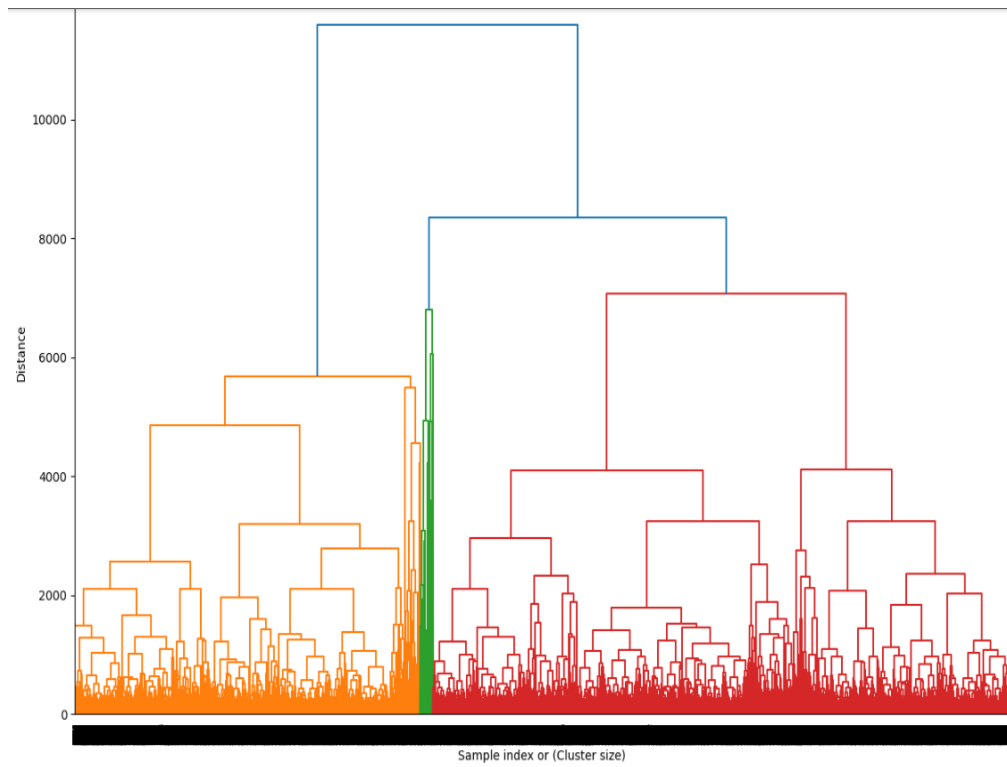


Figure 6: The dendrogram drawn for n_clusters = 4, Customer data size = 30,000 rows.

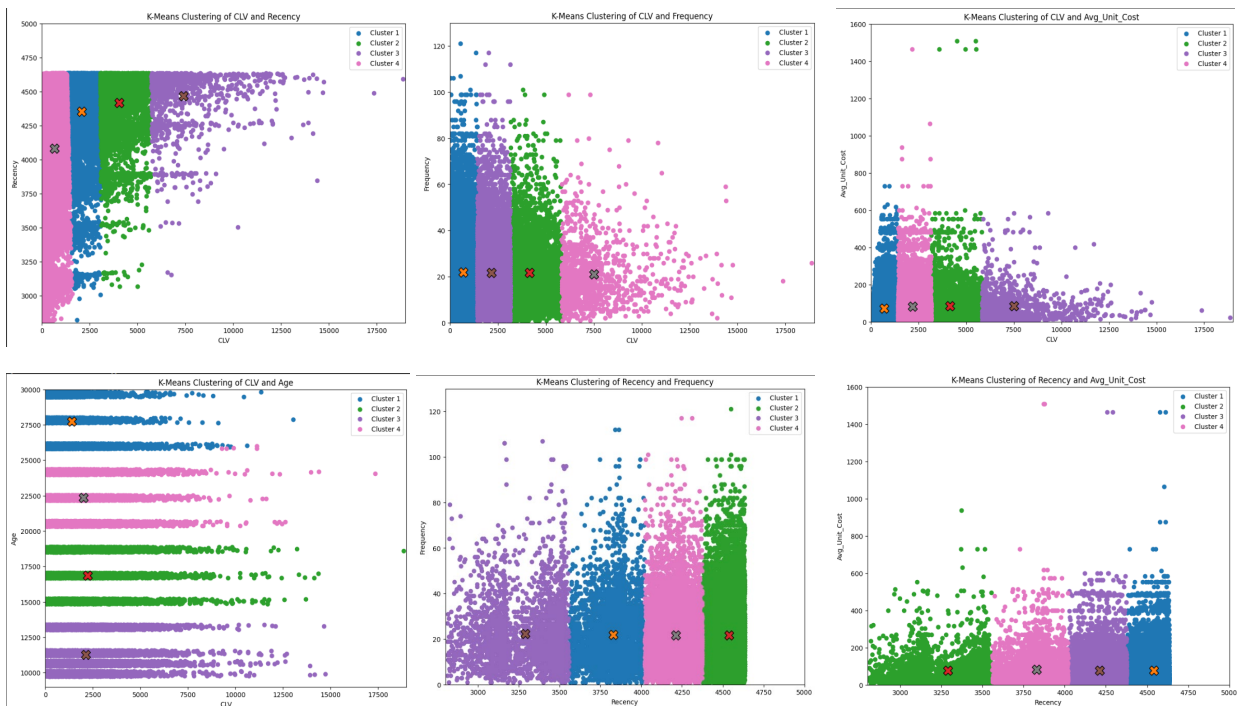


Figure 5a - 6f: Pair-wise visualisation of k-means clustering for features of Customer data.

Figure 5 depicts the resulting dendrogram with 4 clusters with a different colour scheme from the one in Figure 4.

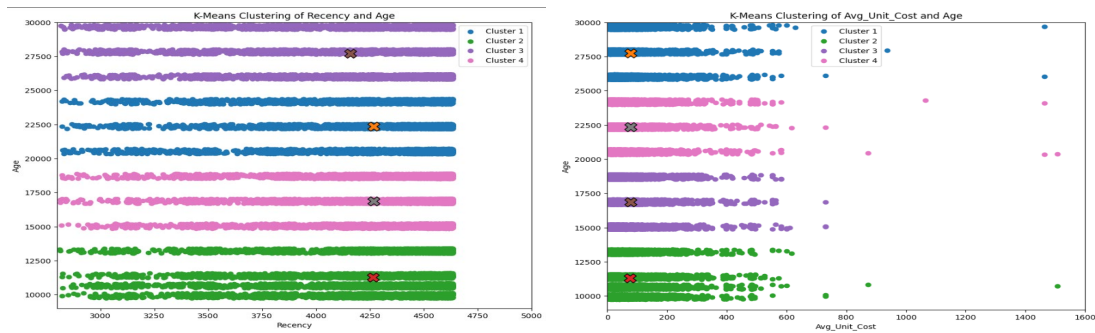


Figure 7a – 7b: Pair-wise visualisation of k-means clustering.

4.4 k-Means Clustering

Figures 6a to 6f, and 7a and 7b show the visualisation of k-means clustering of the sample data for pair-wise features in the dataset. Outliers are clearly present in every cluster. These are also present in the cluster-wise box-plots of the five features shown in the accompanying python notebook for this mini-project.

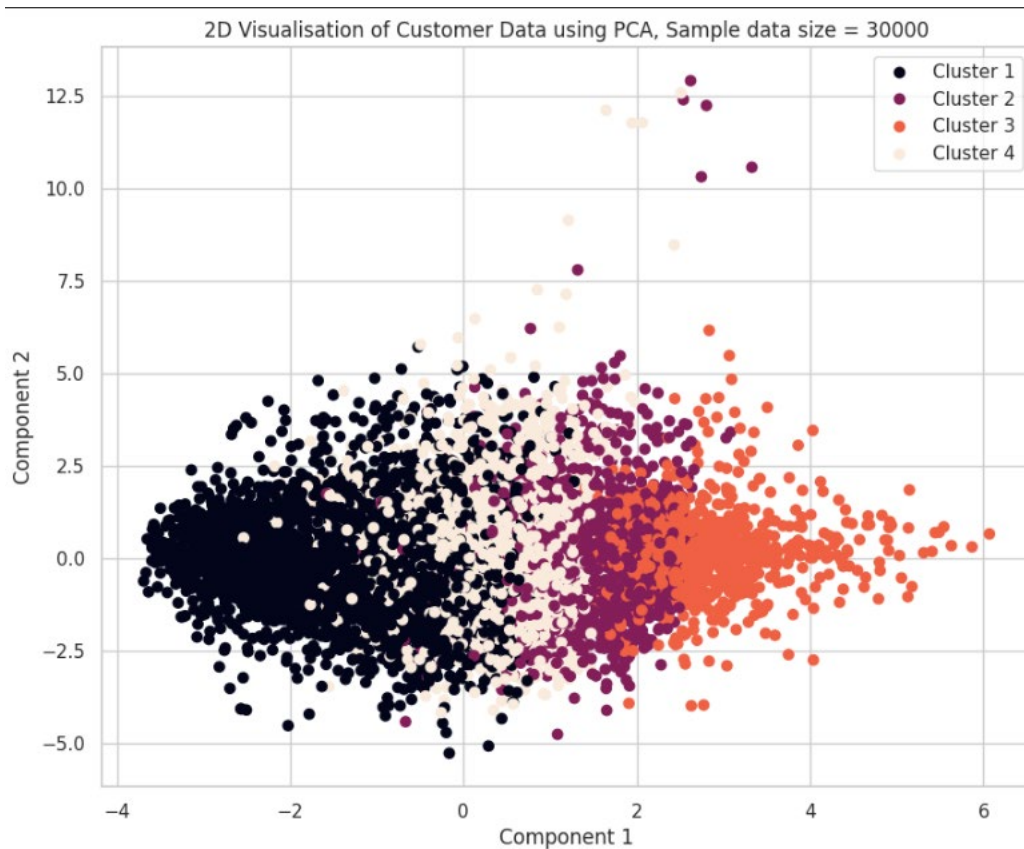


Figure 8: 2D Cluster Visualisation of Customer data using PCA.

4.5 Dimensionality Reduction using PCA and t-SNE

Dimensionality reduction methods, namely: PCA (Figure 7) and t-SNE (Figure 8) provide clearer visualisations of the customer segmentation in these features. The PCA output shows some cluster points overlapping, which could further be explored by increasing the number of clusters, if time allowed.

The visualisation based on the t-SNE method shows more openings (Figure 8) within data points, which means that the PCA method works on the basis of variance within data points within a cluster, hence the cluster points are closer to each other, albeit with outliers (Figure 7).

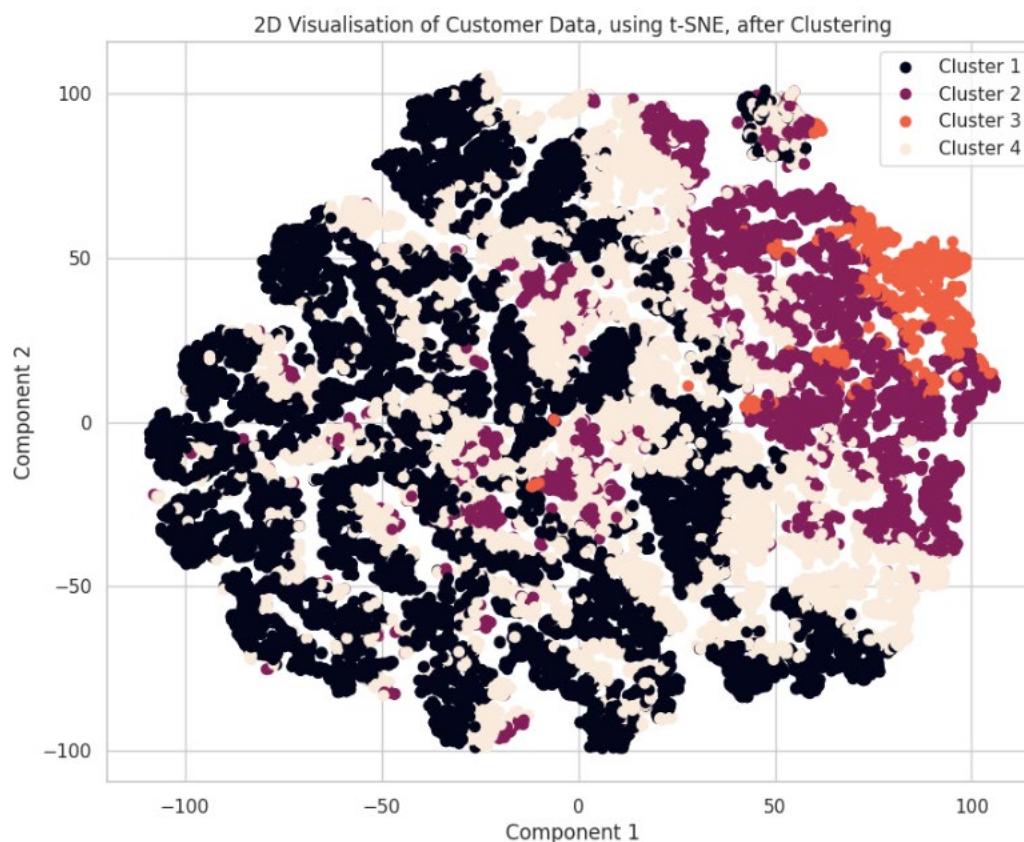


Figure 9: 2D Clusters Visualisation of Customer data, using the t-SNE method.

5. Conclusions

The customer dataset had several rows for every customer which were grouped so that there was one row per customer. This reduced the data size from 951668 to 68299, and the columns were reduced from 20 to merely five. The indication of Silhouette score to have the best value of 2 is, in our view, too low for this large data. So, we selected the number suggested by the Elbow method as 4. For the 2D visualisation, the outputs for PCA and t-SNE methods suggest that we could further explore adjusting the perplexity parameter and observe the variation in the way these clusters are shown.

6. References

Cardona, A. (2024), Applying Statistics and core data science techniques in business: Customer Segmentation with Clustering, provided on Canvas, Course 1, CAM_DS_101 DS Cohort 2, University of Cambridge Institute of Continuing Education (ICE), URL: <https://platform.fourthrev.com/courses/743> .

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